MISH 29
Electrical Safety

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Electrical Safety

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MH CONSTRUCTION INDUSTRIAL SAFETY AND HEALTH PROGRAM

Title

ELECTRICAL SAFETY

Number

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1.0 PURPOSE

To define the requirements for electrical safety and provide specific reference sources for other requirements, and to ensure compliance with national Fire Protection Association-National Electrical Code (NFPA-NEC), and Occupational Safety and Health Administration/Washington Industrial Safety and Health Act (OSHA/WISHA) requirements.

2.0 RESPONSIBILITY

2.1 The Key Supervisor shall be responsible for the following:

• Monitoring those operations involving electrical work.

• Ensuring compliance with the provisions of this procedure, including required training and compliance with permits.

• Ensuring compliance of all applicable Federal, State and applicable codes.

2.2 MH Construction Industrial Safety and Health (IS&H) shall be responsible for reviewing and approving required work permits.

3.0 DEFINITIONS

Ground: A conducting connection, whether intentional or accidental, between an electric circuit or equipment and the earth (or to some conducting body which serves in place of the earth).

Ground-Fault Circuit Interrupter (GCI): A device for the protection of personnel that de-energizes a circuit or portion thereof within an established period of time when current to ground exceeds some predetermined value less than that required to operate the overcurrent protective device of the supply circuit.

4.0 GENERAL REQUIREMENTS

4.1 Electrical work shall be conducted and electrical circuits, equipment, and components shall be installed and maintained in accordance with the following requirements:

• 29 Code of Federal Regulations (CFR) 1910, Subpart S; Electrical

• Washington Administrative Code (WAC) 296-155, Part I; Electrical

• Washington Administrative Code (WAC) 296-44; Washington State Electrical Construction Code

• Washington Administrative Code (WAC) 296-45; Washington State Electrical Work Safety Rules

4.2 WHEN DEEMED APPROPRIATE, THE IS&H DEPARTMENT MAY, AFTER CAREFUL EVALUATION, GRANT EXEMPTION FROM SPECIAL REQUIREMENTS OF THIS PROCEDURE FOR SPECIFIC ELECTRIACAL WORK ON AN INDIVIDUAL BASIS. Justification for granting such exemptions shall be documented on the Energized Electrical Work Permit (attachment MISH 29-A) or on an attachment thereto.

4.3 Each employee shall receive electrical safety training based on his or her potential exposure. Only qualified persons are allowed to work on or near exposed electrical parts.

4.4 Conductors and parts of electrical equipment that have been de-energized but not been locked or tagged out shall be treated as live parts.

4.5 Live parts to which an employee may be exposed shall be de-energized and locked and/or tagged (in accordance with MISH 15) before work is performed on or near them. However, in certain situations, de-energizing introduces additional or increased hazards or is infeasible due to equipment design or operational limitations. In the situations, written justification for conducting work on energized electrical parts and a detailed description of the safety-related work practices to be used to protect employees shall be completed.

Live parts that operate at less than 50 volts to ground need not be de-energized if there will be no increased exposure to electrical burns or to explosion due to electric arcs.

4.5.1 Examples of increased or additional hazards include the following:

• Interruption of life support equipment.

• Deactivation of emergency alarm systems.

• Shutdown of hazardous location ventilation equipment.

• Removal of illumination for an area.

4.5.2 Examples of work that may be performed on or near energized circuit parts due to equipment design or operational limitations include the following:

• Testing of electric circuits that can only be performed with the circuit energized.

• Work on circuits that form an integral part of a critical continuous process that would otherwise need to be completely shut down in order to permit work on one circuit or piece of equipment.

4.5.2.1 Except in special circumstances, normal operations do not constitute a critical continuous process.

Example: While maintaining continuous operation of several computers may be essential to the normal conduct of business, it would not be considered a critical continuous process unless the interruption of electrical power to the computers would result in one of the following:

• A threat to safety or security.

• Loss of critical data.

• Loss of monitoring capability for laboratory/experimental operations.

4.5.2.2 MH Construction IS&H shall be consulted when there are questions as to whether an operation is a critical continuous process.

4.6 When exposed live parts are not de-energized (for reasons of increased hazards or infeasibility), appropriate safety measures shall be documented on the Energized Electrical Work Permit and fully implemented to protect employees who may be exposed to the electrical hazards involved.

4.6.1 Safety-related work practices shall be designed to protect employees against direct contact of energized circuit parts with any part of the body or indirect contact through some other conductive object.

4.6.2 Work practices used shall be suitable for the conditions under which the work is to be performed and for the voltage level of the exposed electric conductors or circuit parts.

4.6.3 Employees working in areas where there are potential electrical hazards shall be provided with, and shall use, electrical protective equipment. The protective equipment shall provide appropriate protection for the specific parts of the body to be protected and for the work to be performed.

4.7 If work will be performed within 20 ft. of overhead utility or electrical lines, the utilities department shall be notified. If the work being performed will approach 10 ft. or closer to overhead electrical lines, the utility department shall be contacted for assistance in de-energizing or insulating the lines.

4.8 Workers shall take special care when handling material in the vicinity of exposed electrical circuits to prevent contact with electrical parts.

4.9 Signs, barricades, and/or attendants shall be used to isolate the work area and warn others of the exposed energized electrical circuits.

4.10 Exposed non-current-carrying metal parts of fixed equipment shall be grounded except when specifically excluded by 29 CRF 1910, Subpart S.

4.11 Exposed non-current-carrying metal parts of cord-and-plug connected equipment shall be grounded.

4.12 Portable power tools shall be either grounded or of a double insulated type. Double insulated tools shall bear a permanent label identifying them as such.

4.13 Double insulated power tools shall be used on energized parts.

4.14 Extension cords shall be grounded, designed for hard or extra-hard usage, and maintained in good repair. Repair to a break in the cord insulation shall be such that the insulating quality of the repair is equal to or grater than the original.

4.15 Electrical disconnect boxes and distribution panels shall be marked as to distribution/purpose.

4.16 Unused openings in electrical boxes and fittings shall be effectively closed.

4.17 Cords shall be protected against damage from traffic, pinch points, and sharp corners. Examples include doorways, roadways, roof edged, etc.

4.18 When working near overhead lines, unqualified persons must maintain a clearance distance of 10'.

4.19 MHC does not have qualified employees to be working in the vicinity of overhead live lines.

4.20 Any vehicle or mechanical equipment capable of having parts of its structure elevated near energized overhead lines shall be operated so that a clearance of 10 ft. (305 cm) is maintained. If the voltage is higher than 50kV, the clearance shall be increased 4 in. (10 cm) for every 10kV over that voltage.

4.21 Employees may not enter spaces containing exposed energized parts unless illumination is provided that enables the employees to work safely.

4.22 Portable ladders shall have non-conductive side rails.

5.0 PROCEDURE

5.1 ENERGIZED ELECTRICAL WORK (50 VOLTS OR MORE):

Prior to work commencing on or near energized electrical parts of 50 volts or more (nominal voltage between any two conductors), the following activities shall be completed:

5.1.1 Measures shall be taken to de-energize all electrical source(s) and ensure that there is no feasible alternative of completing the work in the vicinity of energized parts.

5.1.2 An Energized Electrical Work Permit shall be completed by the Key Supervisor and approved by IS&H. Standardized permits, which are developed and implemented for routine tasks, shall be reviewed periodically (not to exceed 6 months).

5.1.3 Employee representatives shall actively participate in establishing protective measures and completing the permit.

5.1.4 The permit shall be reviewed with the worker(s) involved in the work activities.

5.2 The completed permit shall be available with other pre-job safety planning documents. 5.3

5.4 PERSONAL PROTECTIVE EQUIPMENT:

Workers who perform work on or near energized parts shall use appropriate personal protective equipment. The use of this equipment is evaluated during pre-job safety planning phases of each job and identified on the permit.

5.4.1 Rubber protective equipment (gloves, sleeves, blankets and mats) shall be used as required.

5.4.1.1 Rubber protective equipment (gloves, mats, etc.) shall be maintained in a safe, reliable condition, stored, and used according to the manufacturers recommendation and not used for other than its intended purpose.

5.4.1.2 Equipment to be used shall be inspected before each use to verify the item is in satisfactory condition and has been tested as required.

5.4.1.3 Rubber gloves shall be inspected for holes and air tested before use.

5.4.1.4 Rubber protective equipment shall be inspected, sterilized and tested by an authorized testing organization at six month intervals.

5.4.2 Nonconductive head protection shall be worn whenever there is a danger of head injury from electrical shock or burns due to contact with exposed electrical parts.

5.4.3 Face protection shall be worn whenever there shall be danger of injury to the eyes or face from electric arc, flashes, or electrical explosion.

5.4.4 Conductive clothing and accessories (e.g., rings, watches, bracelets, metal frame glasses, metal hats, etc.) shall be removed.

5.4.5 Natural fiber clothing (e.g., cotton welders coveralls) shall be worn by all employees performing work on energized parts. Clothing shall be inspected for rips, tears, and/or flaws in material. If clothing is damaged in any way, it is not used.

5.4.6 Only voltage rated insulated tools or insulating protective barriers, which are approved by a nationally recognized testing organization, shall be used.

5.4.7 Conductive items of jewelry or clothing shall not be worn unless they are rendered non-conductive by covering, wrapping or other insulating means.

5.5 ELECTICAL TEST EQUIPMENT

5.5.1 Only qualified personnel shall use electrical test equipment.

5.5.2 Electrical test equipment shall be visually inspected immediately before use and not used if defective. Tagging it out of service identifies defective test equipment.

5.5.3 Electrical test equipment shall be used only for intended applications.

5.5.4 When verifying that circuits are de-energized, electrical test equipment shall be checked on a known energized source for proper operation immediately before and after use.

5.5.5 When performing zero energy checks, employees shall assure that stored electrical or mechanical energy cannot re-energize the circuit.

5.6 EQUIPMENT GROUNDING

5.6.1 Cord sets, receptacles, and portable and stationary equipment shall be grounded in accordance with the requirements set forth in the standards identified in subsection 4.1

5.6.2 On construction sites, all cord sets, receptacles, and portable equipment connected by a cord or plug shall be visually inspected prior to use.

5.7 GROUNG-FAULT PROTECTION FOR PERSONNEL:

On construction sites, the Contractor shall provide ground-fault circuit interrupters (GFCI) for all 125-volt, single-phase, 15- and 20-ampere receptacle outlets which are not a part of the permanent wiring of the building or structure. GFCI protection shall cover all cord sets, receptacles which are not part of the permanent wiring of the building or structure, and equipment connected by cord and plug which are available for use by employees.

Extension cords, although plugged into a permanent outlet, shall be considered temporary wiring and shall be used only with ground fault circuit interrupters.

EXCEPTION: Receptacles on a 2-wire single phase portable or vehicle-mounted generator rated not more than 5 KW, where the circuit conductors of the generator are insulated from the generator frame and all other ground surfaces.

5.8 ASSURED GROUNDING PROGRAM

5.8.1 Receptacles and cords more than 125-volts shall be part of the assured grounding program.

5.8.2 A competent person shall inspect all cords above 125-volts on a quarterly basis, before first use, before use after any incident that reasonably could cause damage, and after any repair. The inspection shall include the following:

▪ Test all equipment grounding conductors for continuity. ▪ Test each receptacle or plug to assure the equipment grounding conductor is connected properly. ▪ Visually inspect each cord for damage or possible damage.

5.8.3 Each cord or receptacle tested on a quarterly basis shall be marked with the corresponding color in accordance with the following table:

Quarter Months Color First Jan-March Yellow Second April-June Green Third July-Sept Red Fourth Oct-Dec Blue

5.9 PORTABLE MULTIPLE OUTLET STRIP

5.9.1 Portable multiple outlet strips (also called surge protectors) are permitted for use with computers, video equipment, and other applications where surge protection is desirable. Portable multiple outlet strips shall not be used for refrigerators, microwave ovens, coffee pots, or other applications where permanent wiring is required.

5.9.2 Use and care of portable multiple outlet strips shall be in accordance with the requirements of the NEC.

5.9.3 Only Underwriters laboratories, Inc. (UL)-listed portable multiple outlet strips shall be used,

5.9.4 Portable multiple outlet strips shall be grounded and equipped with internal circuit breaker protection.

5.9.5 Portable multiple outlet strips shall not be affixed to structures, furniture, etc.

5.9.6 Extension cords shall not be used in conjunction with portable multiple outlet strips.

5.9.7 Portable multiple outlet strips shall not be plugged one into another (daisy chained) to accommodate additional appliances/equipment. Where the use of a single portable multiple outlet strip does not provide sufficient quantities of outlets, additional permanent receptacle outlets shall be installed in accordance with applicable codes.

6.0 RECORDS

Document Record Submittal Responsibility Record Retention Responsibility Energized Electrical Work Permit

Contractor Construction Management

7.0 REFERENCES

National Fire Protection Association, National Electric Code (NFPA-70)

8.0 ATTACHMENTS

MISH 29-A Energized Electrical Work Permit

MH CONSTRUCTION ENERGIZED ELECTRICAL WORK PERMIT PR0JECT:_____PROJECT #:_____ LOCATION OF WORK:_____ JOB DESCRIPTION

(INCLUDE ACTIVITIES TO BE PERFORMED, COMPONENTS INVOLVED, VOLTAGES, ETC.)

_____

ENERGIZED ELECTRICAL WORK AUTHORIZATION JUSTIFICATION

( ) INTRODUCES ADDITIONAL HAZARDS ( ) INFEASIBLE DUE TO EQUIPT. DESIGN

( ) INTRODUCES INCREASED HAZARDS ( ) INFEASIBLE DUE TO OPERATIONAL LIMITATIONS EXPLAIN JUSTIFICATION:_____

_____

SAFE WORK REQUIREMENTS

( ) Qualified Personnel assigned to perform work.

( ) Person trained in First-Aid/CPR & emergency electrical rescue procedures, standing by. Standby person to be equipped with insulating equipment needed for rescue.

( ) Adequate working clearance available, escape route(s) in case of emergency.

( ) Communications available and in operating condition. EMERGENCY NUMBER:_____

( ) Work area has adequate lighting.

( ) Work area control measures: ( ) barriers & signs ( ) attendant

( ) Conductive items removed: metal rings, watches, jewelry, buckles, badge holders, etc.

( ) De-energize and isolate equipment to the extent feasible.

( ) Insulating protective equipt. used: ( ) line hoses & covers ( ) rubber insulating matting ( ) rubber blankets

( ) Voltage rated, insulated tools.

( ) Portable ladders with clean & dry, non-conductive side rails.

( ) Testing equipment at work site compatible with voltage present.

( ) Other:_____

PERSONAL PROTECTIVE EQUIPMENT

( ) non-conductive head protection ( ) safety glasses ( ) face shield ( ) rubber gloves/protectors ( ) rubber sleeves ( ) non-conductive clothing ( )other:_____( )other:_____

MH CONSTRUCTION SAFETY SIGNOFF:_____**DATE:**_____

SUPERVISOR(S) SIGNOFF:_____

WORKER(S) SIGNOFF:_____

_____

MISH 29 — Electrical Safety

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